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Neurobloom Software Requirements Specification Documentation



Department of Artificial Intelligence & Data Science

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I. Introduction

NeuroBloom is a technology-driven screening platform built to identify learning disabilities — including dyslexia, ADHD, dysgraphia, and dyscalculia — in school-aged children at the earliest possible stage. The platform weaves together gamified diagnostic assessments, behavioral data analytics, and practitioner connectivity to give schools and families a reliable, stress-free way to detect cognitive difficulties before they compound.

Research consistently shows that roughly two out of every ten students below the age of fifteen experience some form of learning difficulty. A significant proportion of these children go unidentified until declining academic performance forces attention to the issue — often years after effective intervention would have been most beneficial.

NeuroBloom resolves this by offering a single, centralized digital environment that connects schools, parents, and qualified practitioners. Students engage with theme-based games specifically engineered to surface cognitive and behavioral signals; an intelligent analytics engine processes those signals in the background; and the resulting reports guide further professional assessment where warranted.

By bridging education, healthcare, and technology within one cohesive product, the platform enables timely intervention — improving academic trajectories, protecting mental health, and helping practitioners grow their patient base organically through verified referrals.

II. Problem Statement

Learning disabilities in children represent a growing, largely invisible public health concern. Despite affecting millions of students, the vast majority of cases remain undiagnosed because accessible, school-friendly screening infrastructure simply does not exist at scale. The downstream consequences — academic failure, eroded self-esteem, anxiety, and social withdrawal — are well-documented and preventable.

Core Problems Identified

1. Absence of Early Detection Mechanisms

Most children with learning disabilities are not identified during the formative years of schooling. Without a diagnosis, they struggle silently, often attributed to laziness or low intelligence rather than a neurological difference.

2. Limited Awareness Among Stakeholders

Parents and teachers frequently misinterpret the signs of dyslexia, ADHD, or dyscalculia as behavioral issues or general academic underperformance, leading to inappropriate responses that worsen outcomes.

3. Fragmented Practitioner Access

Psychologists and therapists who specialize in learning disabilities are scattered and difficult to locate. Ironically, thousands of qualified practitioners struggle to maintain viable practices due to low patient volume — a supply-demand mismatch driven by poor discovery infrastructure.

4. Compounding Academic and Psychological Harm

Undiagnosed conditions deteriorate over time. Affected students are disproportionately prone to depression, anxiety, low self-esteem, and social isolation — outcomes that extend well beyond the classroom.

5. No Integrated Platform Exists

Current digital health tools offer teleconsultation or general mental-health support. None provide early-stage, school-integrated screening specifically designed around learning disabilities.

III. Project Objectives

The Neuro-Bloom platform is guided by the following core objectives:

- Enable early identification of learning disabilities through gamified behavioral screening and intelligent data analysis.
- Provide schools with a deployment-ready platform that requires no specialist hardware or teacher training.
- Improve student outcomes by facilitating timely professional diagnosis and intervention.
- Create a unified environment where parents, schools, and practitioners can collaborate around a child's learning profile.
- Leverage behavioral data from gameplay to generate accurate, low-stress screening reports.
- Raise awareness among parents and educators through embedded educational resources.

- Architect the system for mass-market scalability across thousands of schools from a single codebase.

IV. Software Process Framework

The Software Process Framework defines how a software system is planned, designed, built, and delivered through a set of structured activities. For Neuro-Bloom, this framework was applied to coordinate multiple stakeholders — students, parents, schools, and practitioners — across a technically complex, multi-module product.

Framework Activities

1. Communication

Stakeholder engagement formed the foundation of the project. The team conducted structured surveys and interviews with parents, educators, psychologists, and therapists to map the real-world friction around learning-disability detection. The primary insight: a centralized, school-deployable platform was needed to bridge the gap between detection and treatment.

2. Planning

Planning activities established the project scope, technology stack, target stakeholders, and go-to-market approach. The platform was conceived as a B2B2C model — schools pay for deployment, students use the screening tools, and practitioners access a patient pipeline through the referral system. Key planning outputs included the gamified test design brief, the behavioral analytics architecture, and role-specific dashboard specifications.

3. Modeling

System modeling produced three primary design artifacts: (i) the overall system architecture diagram illustrating the five-layer technical structure; (ii) a database schema covering student profiles, test results, behavioral metrics, and practitioner records; and (iii) UI/UX wireframes for each user role — student, parent, school administrator, and practitioner.

4. Construction

Development proceeded module by module: the web application backend, gamified screening test logic, behavioral analytics engine, user authentication, and automated reporting system. Testing covered functionality, usability, data validation, and end-to-end integration across all user roles.

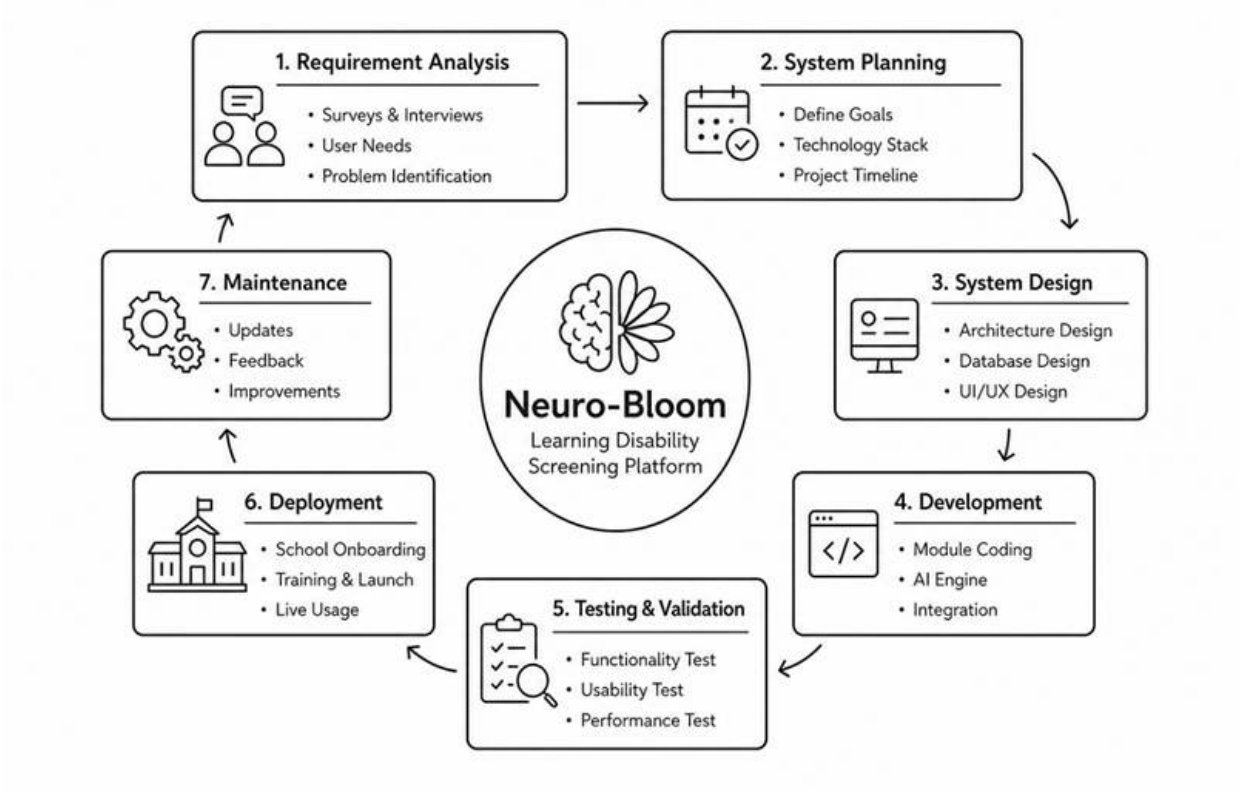
5. Deployment

Deployment entails school onboarding, administrator training, live student screening, and automated report delivery to parents. Because the platform runs on existing school hardware, large-scale rollout

requires no additional infrastructure investment. Continuous post-deployment feedback loops feed back into sprint planning for the next release cycle.

V. Platform Lifecycle

The lifecycle of the Neuro-Bloom platform progresses through seven sequential yet iterative stages:



Stage	Description
1 — Requirement Analysis	Identify the problem landscape through stakeholder surveys, academic research, and competitor analysis.
2 — System Planning	Define platform goals, stakeholder roles, architectural approach, and technology selection.
3 — System Design	Produce data-flow diagrams, module specifications, database schemas, and UX wireframes.
4 — Development	Build screening tests, analytics engine, dashboards, and API integrations sprint by sprint.
5 — Testing	Verify functionality, usability, screening accuracy, and performance under simulated

	load.
6 — Deployment	Onboard pilot schools, train administrators, and release the platform to live users.
7 — Maintenance	Collect feedback, retrain ML models, release patches, and plan feature expansions.

VI. Software Models and Selection Rationale

Three industry-standard software development models were evaluated for suitability against the Neuro-Bloom project's characteristics:

Waterfall Model

A strictly sequential approach where each phase — requirements, design, development, testing, deployment — must be completed before the next begins. While straightforward to manage, it is poorly suited to projects with evolving requirements or iterative discovery needs.

Incremental Model

The system is partitioned into independent modules that are designed, built, and delivered one at a time. This allows partial deployment earlier than a full-system release, but lacks the cross-functional feedback loops necessary for a data-driven platform like Neuro-Bloom.

Agile Model {Selected}

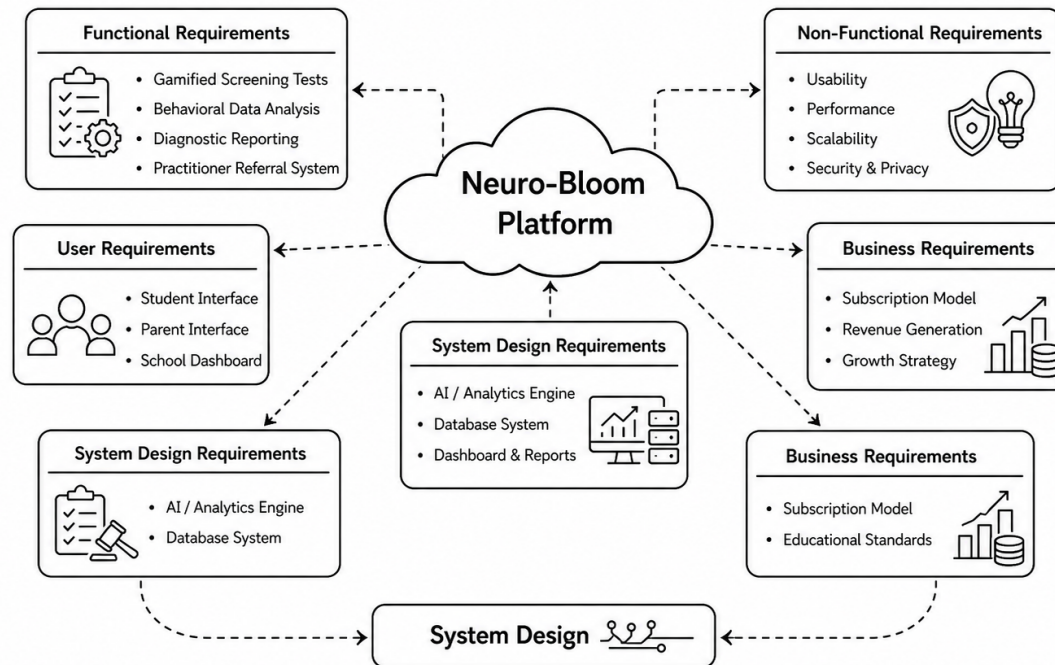
Agile divides development into short, time-boxed iterations called sprints. Each sprint produces a working, tested increment of the product. Cross-functional teams collaborate daily; stakeholders review outputs at the end of each sprint and provide feedback that shapes the next cycle.

Why Agile is the Right Fit for Neuro-Bloom

- The behavioral analysis models require continuous retraining as new screening data accumulates — a pattern that maps naturally to iterative sprints.
- Feedback from schools, parents, and practitioners is essential for calibrating the platform; Agile's sprint-review cadence formalizes this loop.
- The platform's distinct modules — screening tests, analytics engine, dashboards, referral system — can be developed in parallel by small, focused teams.
- Pilot deployments in Pune schools can begin with a minimal feature set, generating real-world data before the full product is complete.
- If a screening algorithm underperforms, Agile allows the team to reprioritize and address the issue in the next sprint without derailing the broader roadmap.

VII. System Requirement Gathering

Neuro-Bloom – System Requirements



Requirements for the Neuro-Bloom platform were collected through a structured, multi-channel research effort:

- Structured surveys and one-on-one interviews conducted with parents and students to understand lived experience of learning difficulties.
- Focus group discussions with psychologists, therapists, and special-education practitioners regarding diagnostic workflow gaps.
- Competitive analysis of existing platforms — including Practo, Wysa, YourDOST, and MediBuddy — to map unaddressed needs.
- Academic literature review covering current diagnostic criteria for dyslexia, ADHD, dysgraphia, and dyscalculia.

The research confirmed a clear gap: parents and schools lack structured, affordable tools for early detection, and existing digital health platforms do not serve the school-integrated, learning-disability-specific use case.

VIII. Functional Requirements

The following table summarizes the primary functional requirements of the Neuro-Bloom platform:

#	Requirement	Description
FR-01	Student Screening	Students access gamified diagnostic tests calibrated to surface cognitive and behavioral signals associated with specific learning disabilities.
FR-02	Behavioral Data Analysis	The system captures gameplay interactions — response latency, error patterns, attention drift — and passes them through ML models for disability likelihood scoring.
FR-03	Diagnostic Report Generation	Upon test completion, an automated report is produced summarizing potential areas of difficulty and recommended next steps.
FR-04	Parent Interface	Parents can view screening results, track longitudinal progress, and receive curated practitioner recommendations.
FR-05	School Dashboard	Administrators can manage cohort screenings, monitor aggregate results, and flag students requiring referral — all without clinical expertise.
FR-06	Practitioner Network	A verified directory of psychologists and therapists is maintained; parents and schools can discover and contact practitioners through the platform.
FR-07	Consultation Booking	Parents can schedule in-platform consultations with network practitioners, creating a closed referral loop.
FR-08	Awareness Resources	Educational content about learning disabilities — causes, symptoms, interventions — is surfaced contextually for parents and teachers.

IX. Non-Functional Requirements

Non-functional requirements define the quality attributes the system must satisfy, independent of specific features:

Attribute	Requirement Detail
Usability	Interfaces must be intuitive for children (ages 6–15), non-technical teachers, and parents unfamiliar with medical terminology. Gamified UX eliminates test-anxiety artifacts.
Performance	Behavioral data processing and report generation must complete within acceptable latency during concurrent school-wide screening sessions.
Scalability	The architecture must support simultaneous use by thousands of students across multiple schools, with horizontal scaling on cloud infrastructure.
Security & Privacy	All student data — including diagnostic results and health records — must be encrypted at rest and in transit, compliant with applicable data protection regulations.

Reliability	The platform must maintain high availability during scheduled school screening windows, with graceful error recovery and no data loss.
Compatibility	The system must run on commodity school hardware (existing desktops, tablets, Chromebooks) using a standard web browser — no installation required.
Maintainability	The codebase must be modular to allow independent updates to ML models, screening content, and UI components without service interruption.

X. System Design Requirements

The system design requirements specify the technical architecture underpinning all platform features. Neuro-Bloom is structured as a five-layer architecture:

1. User Interface Layer

Four distinct interfaces serve the platform's user roles: a game-based screening environment for students; a report and consultation hub for parents; a management and analytics dashboard for school administrators; and a consultation management portal for practitioners.

2. Application Logic Layer

The business logic layer orchestrates all core operations: test session management, behavioral data collection and normalization, automated report generation, practitioner recommendations, and user authentication with role-based access control.

3. AI / Behavioral Analysis Engine

This is the analytical core of the platform. Gameplay interaction data — click timing, error frequency, hesitation patterns, and navigational behavior — is processed by purpose-trained machine learning models (Random Forest, SVM, CNN, and LLM-based reasoning, depending on the disability type) to compute screening probability scores.

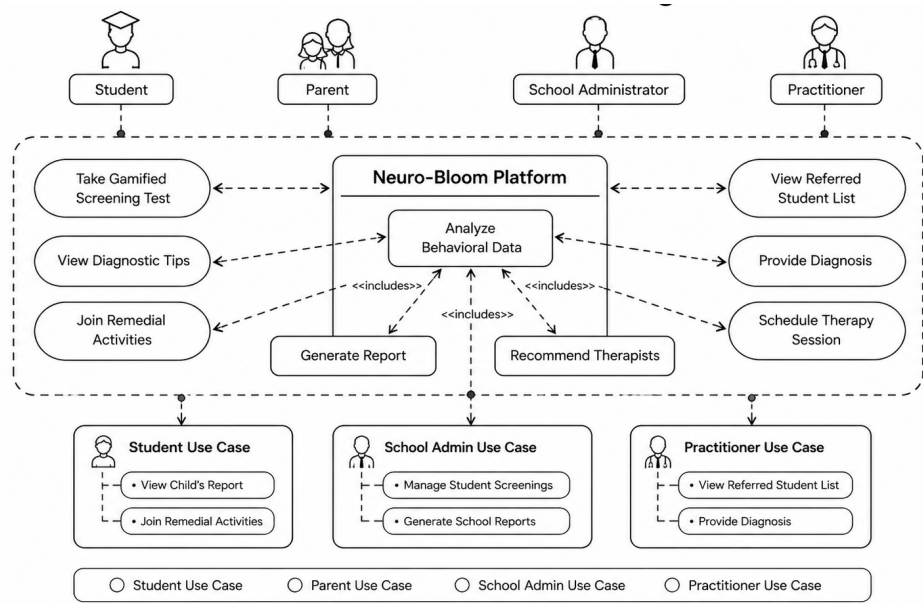
4. Database Layer

The data persistence layer stores student profiles, screening session records, raw behavioral metrics, diagnostic reports, practitioner credentials, and school metadata. Data isolation between schools is enforced at the application layer.

5. Integration Layer

External integrations connect the platform to school information systems, practitioner scheduling tools, and future partners. An API gateway manages authentication and rate limiting for all external consumers.

XI. UML Design



1. Use Case Diagram

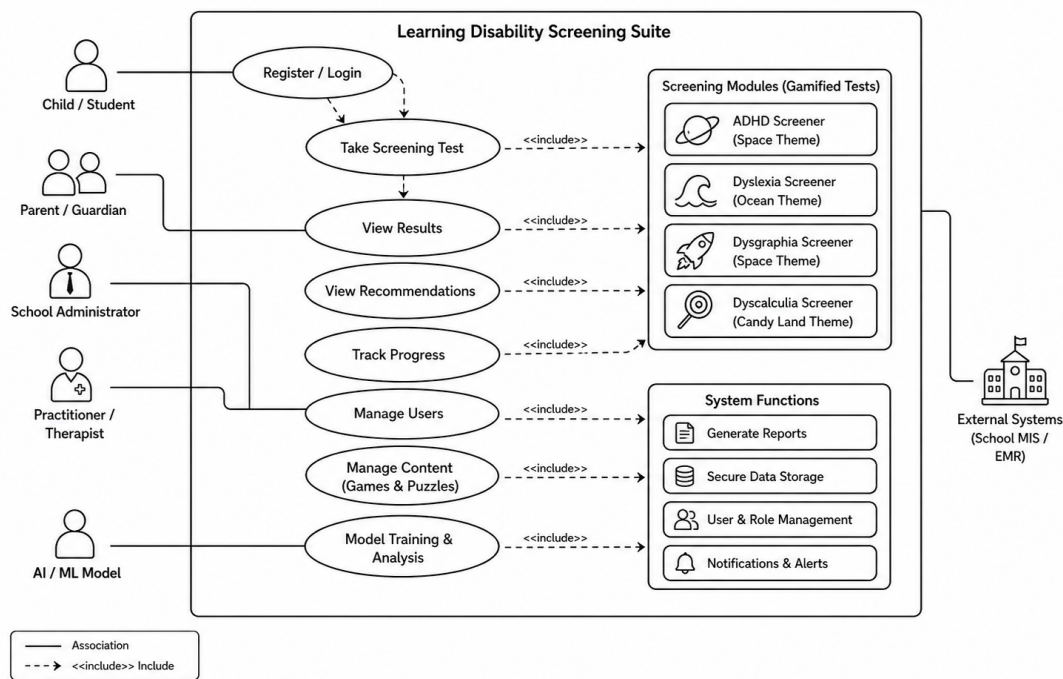


Figure 1: Use Case Diagram illustrating interactions between primary actors (Student, Parent, School Administrator, Practitioner) and the core system functions.

2. Class Diagram

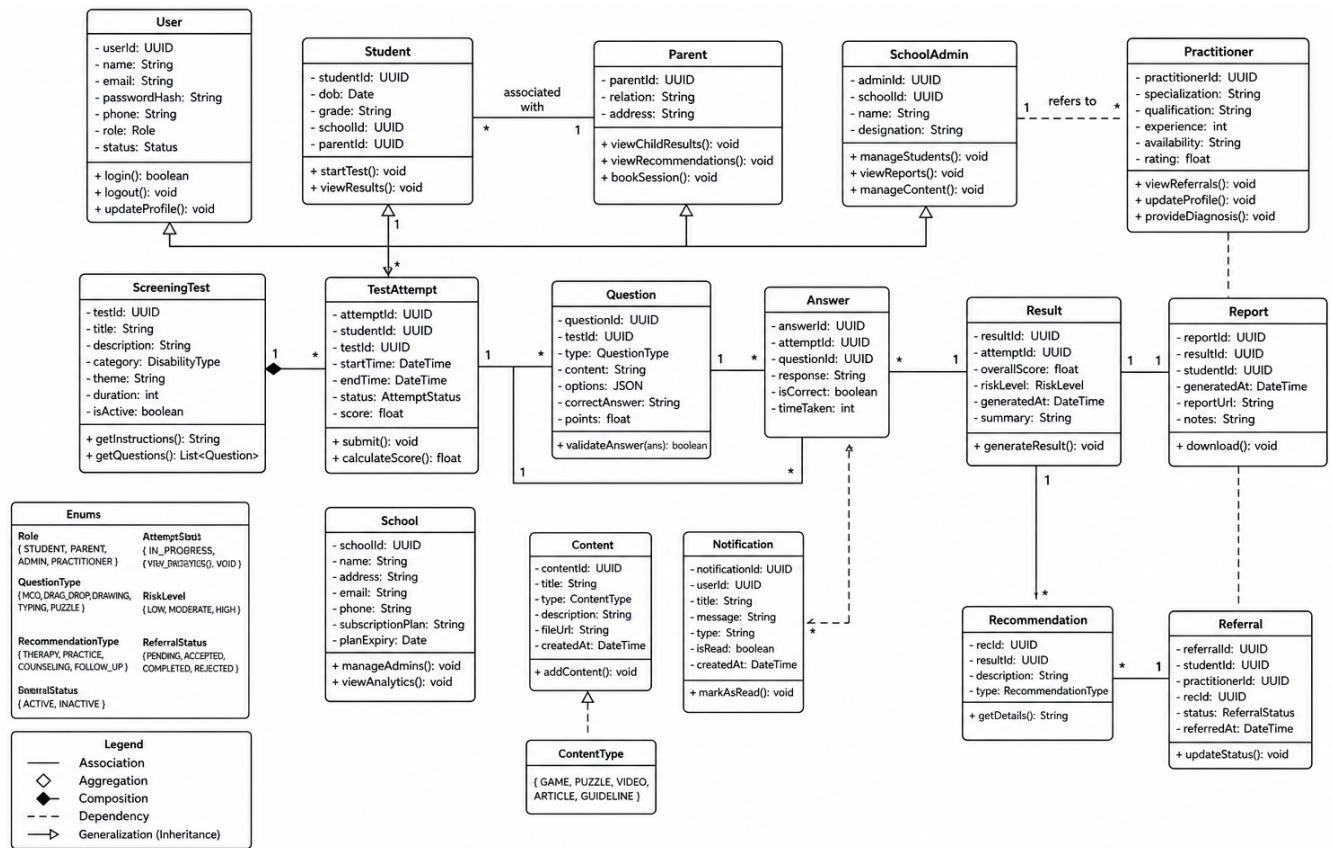


Figure 2: Class Diagram representing the primary entities — User, Student, Test, Report, Practitioner, School — and their relationships within the system data model.

3. Activity Diagram

Learning Disability Screening Suite – Activity Diagram

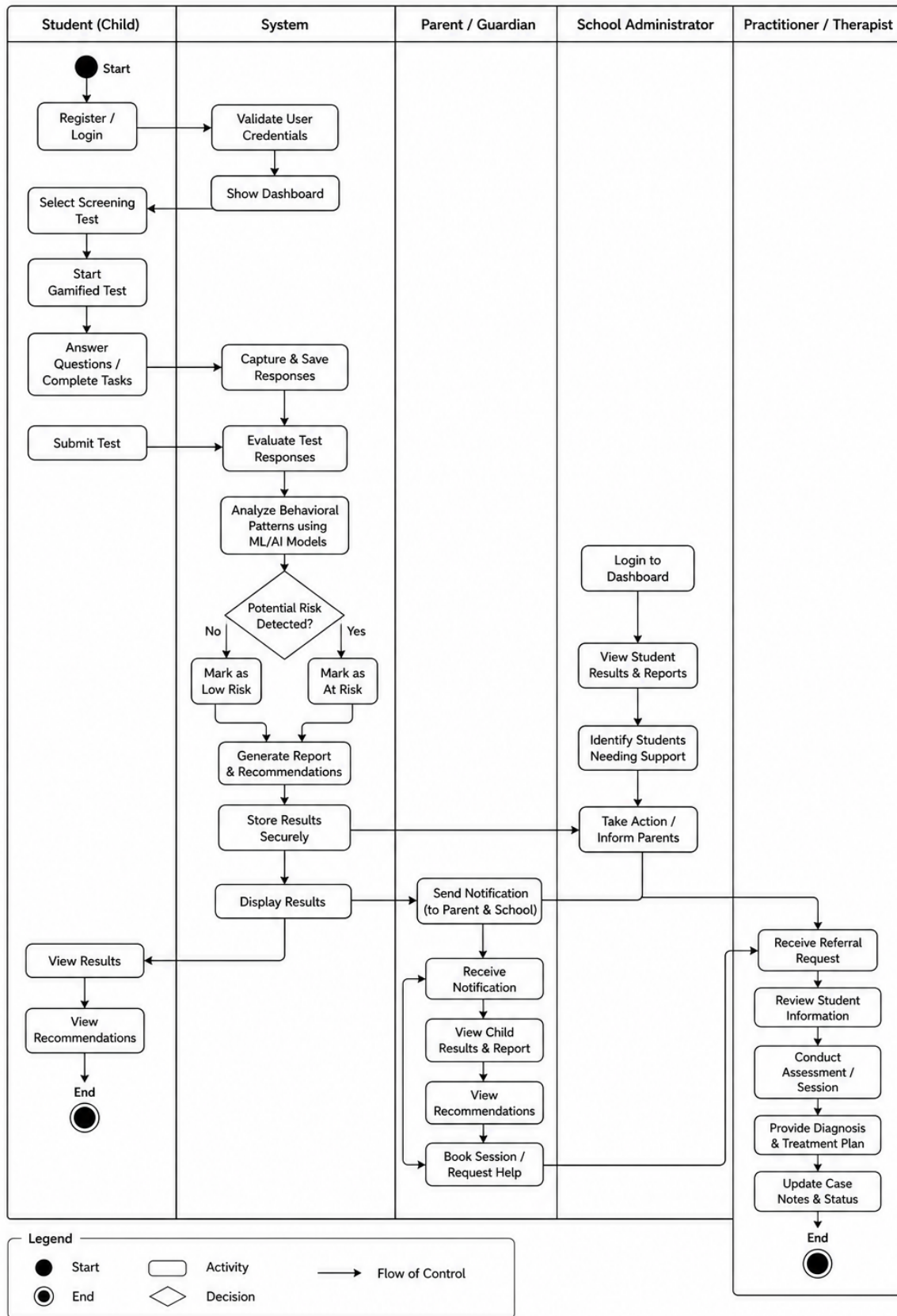


Figure 3: Activity Diagram depicting the end-to-end workflow from student test initiation through behavioral data processing, report generation, and optional practitioner referral.

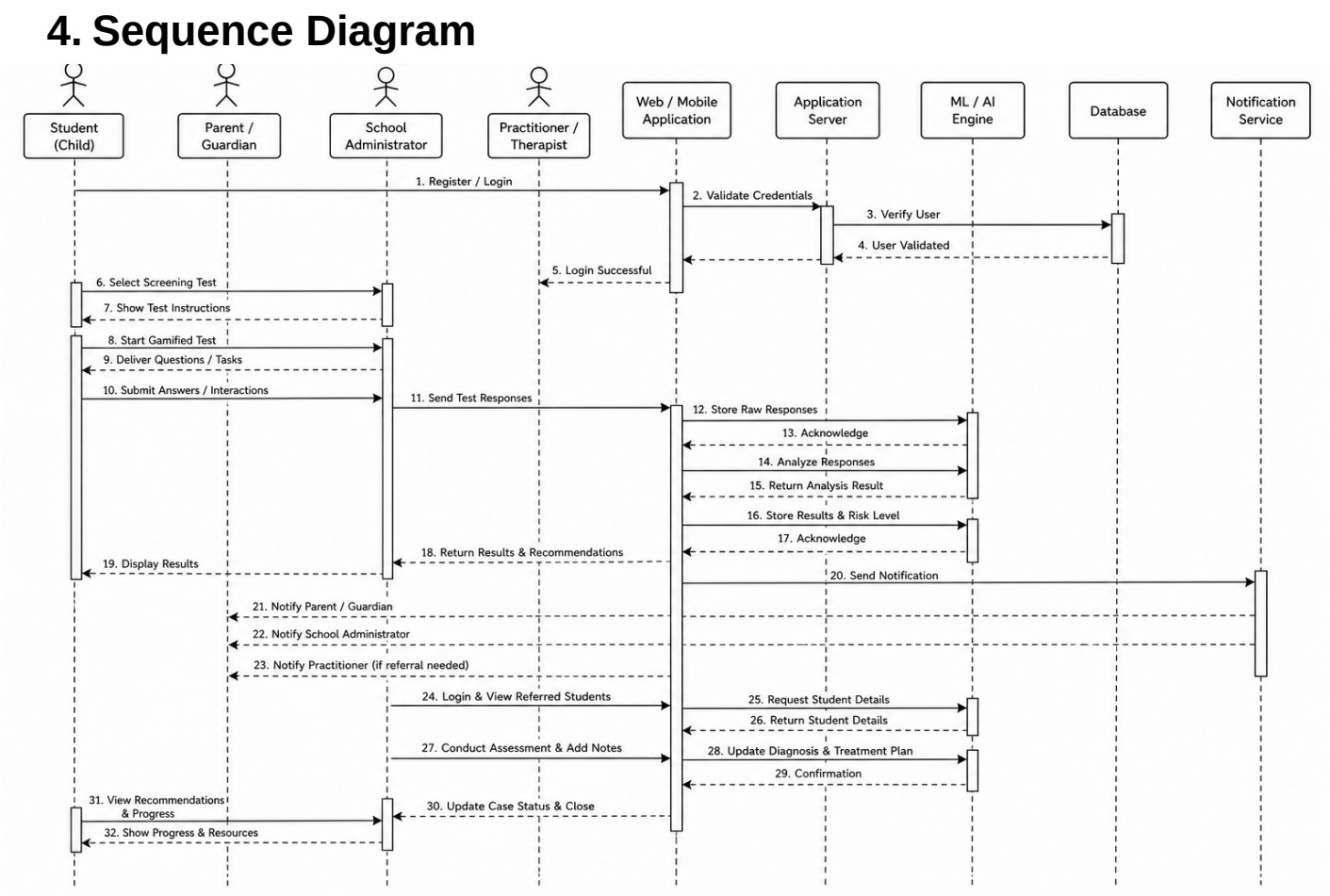


Figure 4: Interaction Sequence Diagram showing the message exchange between system components during a standard screening session.

XII. Key Platform Features

Feature	Detail
Gamified LD Screening	Theme-based interactive games (Space, Ocean, Candy Land) replace clinical questionnaires. Children engage as players; the system observes as a diagnostician. Anxiety is minimized; data quality is improved.
Multi-Disorder Coverage	Four independently deployable screeners target ADHD (Random Forest), Dyslexia (Hybrid SVM + RF + Gradient Boost), Dysgraphia (CNN + image processing), and Dyscalculia (LLM-based reasoning).
Behavioral Analytics Engine	Response latency, error sequences, attention drift, and interaction patterns are captured and analyzed in real time, producing probability scores for each condition.
Automated Diagnostic Reports	Upon completion, the platform generates structured reports summarizing likely difficulty areas and recommending whether professional assessment is warranted.
School Management Dashboard	Principals and administrators gain cohort-level visibility — screening completion rates, risk distribution, and individual flags — without any clinical training.
Parent Monitoring Interface	Parents access their child's longitudinal screening history, progress notes, and curated practitioner referral recommendations through a secure portal.
Practitioner Referral Network	Verified psychologists and therapists are discoverable through the platform, creating a measurable patient pipeline and closing the referral loop.
Mass-Market Scalability	The platform runs on existing school devices via browser, requires no hardware investment, and is designed for low-cost deployment at thousands of schools.

XIII. System Architecture

The Neuro-Bloom system architecture follows a layered, service-oriented design that separates concerns across five distinct tiers, enabling independent scaling and maintenance of each component:

Layer	Components & Responsibilities
Presentation Layer	Student game interface · Parent dashboard · School admin console · Practitioner portal · Responsive web application (Next.js + Tailwind CSS)
Application Layer	Screening test execution engine · Behavioral data collection pipeline · Report generation service · Practitioner recommendation logic · User authentication and RBAC
AI / Analytics Layer	ADHD screener: Random Forest classifier · Dyslexia screener: SVM + RF + Gradient Boost ensemble · Dysgraphia screener: CNN + OpenCV · Dyscalculia screener: LLM reasoning engine
Database Layer	Student profiles · Test session records · Behavioral metrics · Diagnostic reports ·

	Practitioner credentials · School metadata
Integration Layer	School information system connectors · Practitioner scheduling APIs · Analytics export hooks · Future mobile and third-party integrations

XIV. Technology Stack

Domain	Technologies
Frontend	Next.js · React · Tailwind CSS
Backend	Node.js · Express.js · REST APIs
Machine Learning	Scikit-learn (Random Forest, SVM, Gradient Boost) · TensorFlow / Keras (CNN) · OpenCV (image processing)
Generative AI	OpenAI API — LLM-based reasoning for dyscalculia screening
Database	PostgreSQL / MongoDB (TBD based on data structure requirements)
Deployment	Vercel (frontend) · Railway / Render (backend) · Cloud object storage (assets)
DevOps	GitHub · CI/CD pipelines · Environment-based configuration management

XV. Expected Outcomes and Future Work

Expected Outcomes

- Early Detection at Scale — Systematic, school-wide screening catches learning disabilities years before academic failure forces the issue.
- Improved Academic Trajectories — Timely diagnosis enables targeted interventions that measurably improve reading, numeracy, and writing outcomes.
- Elevated Stakeholder Awareness — Embedded educational resources shift parent and teacher attitudes from misattribution to informed support.
- Efficient Practitioner Access — Verified referrals reduce the time between diagnosis suspicion and professional evaluation from months to days.
- Revenue-Generating Pilot in Pune — 15,000–25,000 students screened annually within the first operational year, validating the B2B2C model.

Future Work

1. Eye-Tracking Integration

Coupling the platform with low-cost eye-tracking peripherals would enable passive analysis of reading saccades and fixation patterns, substantially improving dyslexia detection sensitivity.

2. Mobile Application Development

A native iOS and Android application would extend access to parents and practitioners beyond the school environment and enable longitudinal monitoring at home.

3. Personalized Learning Programs

Future versions could generate individualized cognitive support plans — curated reading strategies, attention-training exercises, and math scaffolding — derived directly from each student's screening profile.

4. Government and NGO Partnerships

Integrations with NGOSCSR programs and government education initiatives could subsidize deployment in public schools, dramatically expanding reach and social impact.

5. Multilingual Support

Expanding the screening content and UI into regional Indian languages would make the platform accessible to students across linguistic demographics.